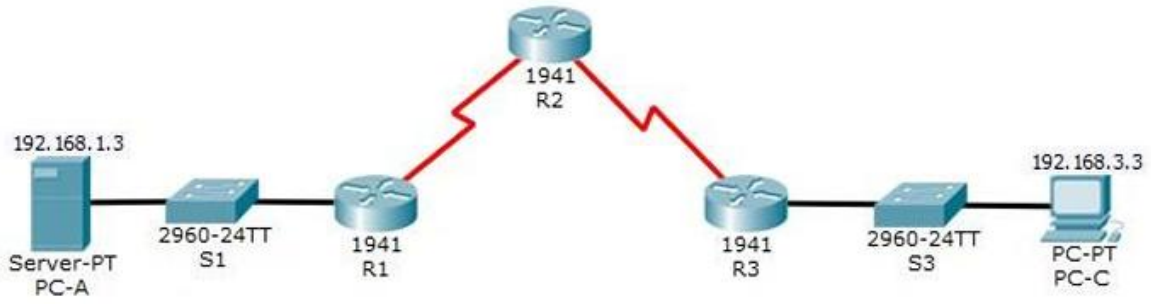
Practical 6: Configuring a Zone-Based Policy Firewall (ZPF) Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Verify connectivity among devices before firewall configuration.
- Configure a zone-based policy (ZPF) firewall on R3.
- Verify ZPF firewall functionality using ping, SSH, and a web browser.

Routing:

Router 1:

Router(config)#hostname R1
R1(config)#Router rip
R1(config-router)#Version 2
R1(config-router)#Network 192.168.1.0
R1(config-router)#Network 10.1.1.0

Router 2:

Router(config)#hostname R2
R2(config)#Router rip
R2(config-router)#Version 2
R2(config-router)#Network 10.2.2.0
R2(config-router)#Network 10.1.1.0

Router 3:

Router(config)#hostname R3
R3(config)#Router rip
R3(config-router)#Version 2
R3(config-router)#Network 192.168.3.0
R3(config-router)#Network 10.2.2.0

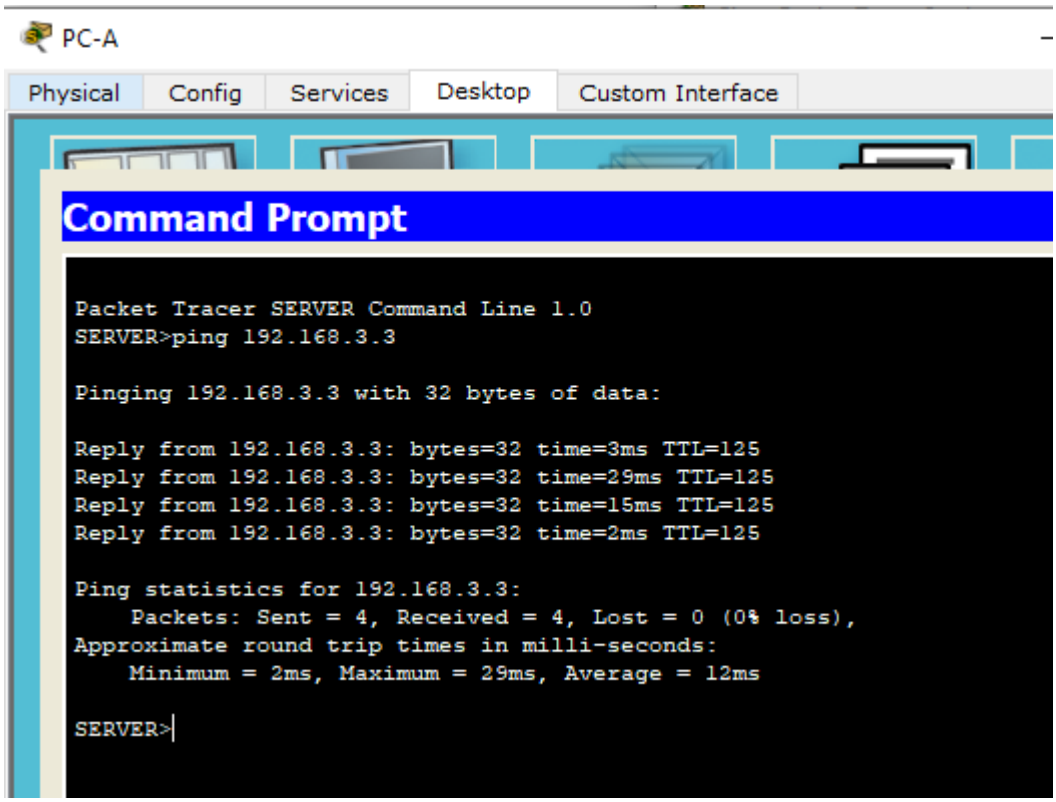
SSH configuration on R2

R2(config)#ip domain-name ccnasecurity.com
R2(config)#username SSHadmin privilege 15 secret ciscosshpa55
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#transport input ssh
R2(config-line)#crypto key generate rsa
The name for the keys will be: R2.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.

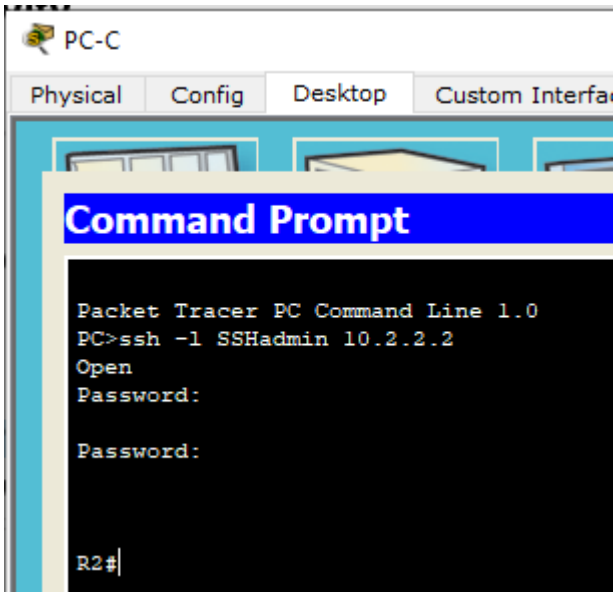
How many bits in the modulus [512]: 1024

Part 1: Verify Basic Network Connectivity

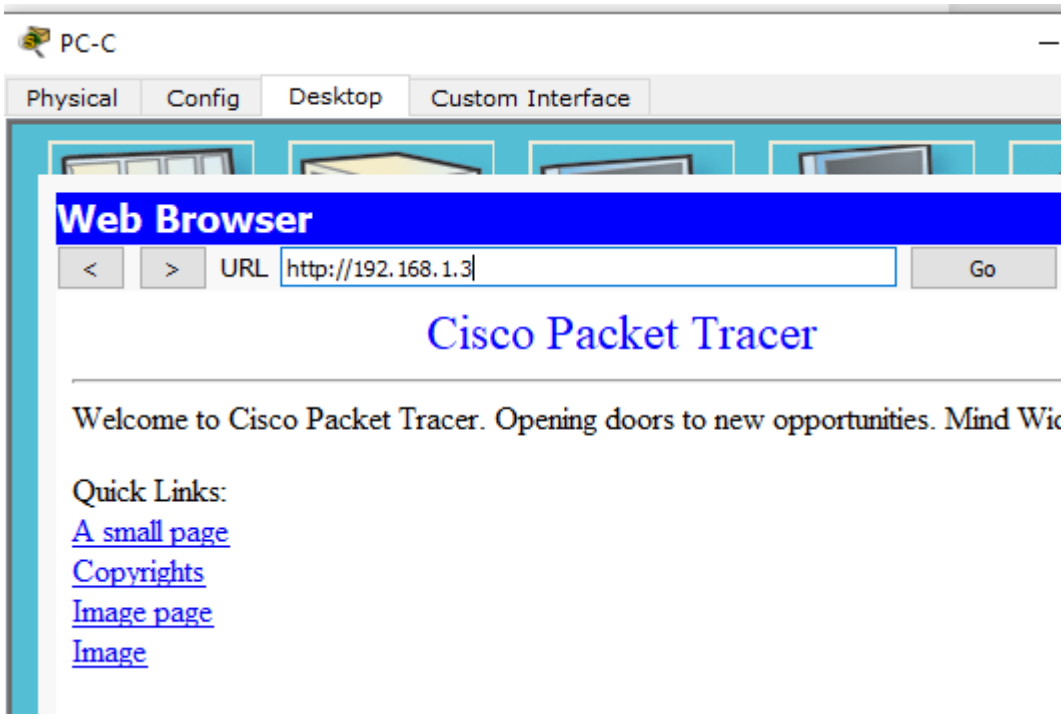
Step 1: From the PC-A command prompt, ping PC-C at 192.168.3.3.



Step 2: Access R2 using SSH.



Step 3: From PC-C, open a web browser to the PC-A server.



Part 2: Create the Firewall Zones on R3

Step 1: Enable the Security Technology package.

a. On R3, issue the show version command to view the Technology Package license information.

R3#show version

```
License Info:

License UDI:

-------------------------------------------------
Device#    PID                    SN
-------------------------------------------------
*0         CISCO1941/K9           FTX1524P70E


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      None          None          None
data          None          None          None

Configuration register is 0x2102
```

b. If the Security Technology package has not been enabled, use the following command to enable the package.

R3(config)#license boot module c1900 technology-package securityk9

c. Accept the end-user license agreement.

d. Save the running-config and reload the router to enable the security license.

```
R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Press enter for destination filename

```
R3#reload
Proceed with reload? [confirm]y
```

press y

e. Verify that the Security Technology package has been enabled by using the show version command.

R3#show version

```
License Info:

License UDI:

-------------------------------------------------
Device#    PID                    SN
-------------------------------------------------
*0         CISCO1941/K9           FTX1524P70E


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          None          None          None

Configuration register is 0x2102
```

Step 2: Create an internal zone. Use the zone security command

R3(config)#zone security IN-ZONE
R3(config-sec-zone)#exit

Step 3: Create an external zone.

R3(config)#zone security OUT-ZONE
R3(config-sec-zone)#exit

Part 3: Identify Traffic Using a Class-Map

Step 1: Create an ACL that defines internal traffic.

R3(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 any

Step 2: Create a class map referencing the internal traffic ACL.

R3(config)# class-map type inspect match-all IN-NET-CLASS-MAP
R3(config-cmap)#match access-group 101
R3(config-cmap)#exit

Part 4: Specify Firewall Policies

Step 1: Create a policy map to determine what to do with matched traffic.

R3(config)# policy-map type inspect IN-2-OUT-PMAP

Step 2: Specify a class type of inspect and reference class map IN-NET-CLASS-MAP.

R3(config-pmap)# class type inspect IN-NET-CLASS-MAP

Step 3: Specify the action of inspect for this policy map.

R3(config-pmap-c)#inspect

R3(config-pmap-c)#exit

R3(config-pmap)#exit

```
R3(config)# policy-map type inspect IN-2-OUT-PMAP
R3(config-pmap)# class type inspect IN-NET-CLASS-MAP
R3(config-pmap-c)#inspect
%No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All
protocols will be inspected
R3(config-pmap-c)#exit
R3(config-pmap)#exit
```

Part 5: Apply Firewall Policies

Step 1: Create a pair of zones.

R3(config)#zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE

Step 2: Specify the policy map for handling the traffic between the two zones.

R3(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)#exit

Step 3: Assign interfaces to the appropriate security zones.

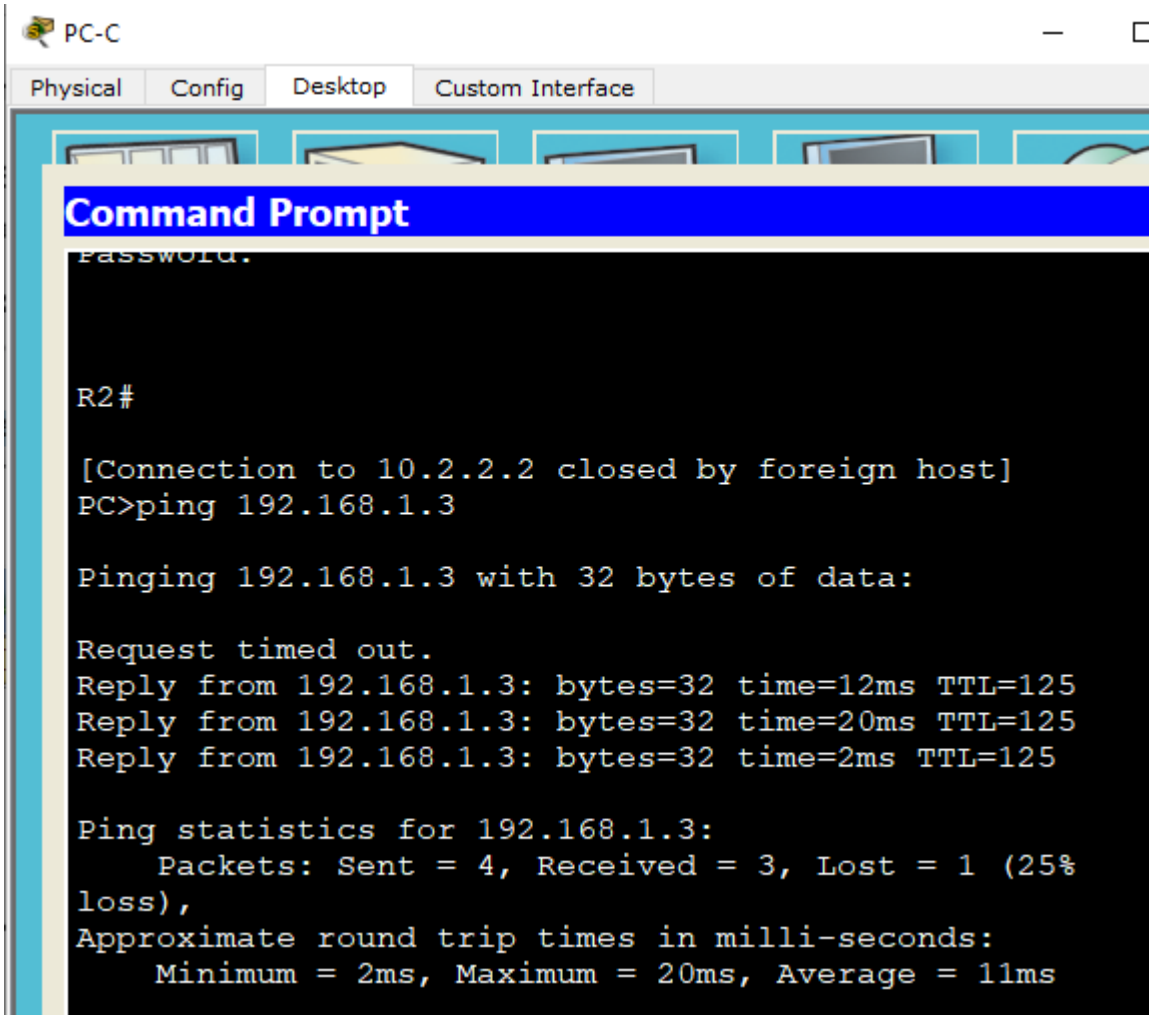
R3(config)#interface g0/1
R3(config-if)#zone-member security IN-ZONE
R3(config-if)#exit
R3(config)#interface s0/0/1
R3(config-if)#zone-member security OUT-ZONE
R3(config-if)#exit

Step 4: Copy the running configuration to the startup configuration.

R3#copy running-config startup-config

Part 6: Test Firewall Functionality from IN-ZONE to OUT-ZONE

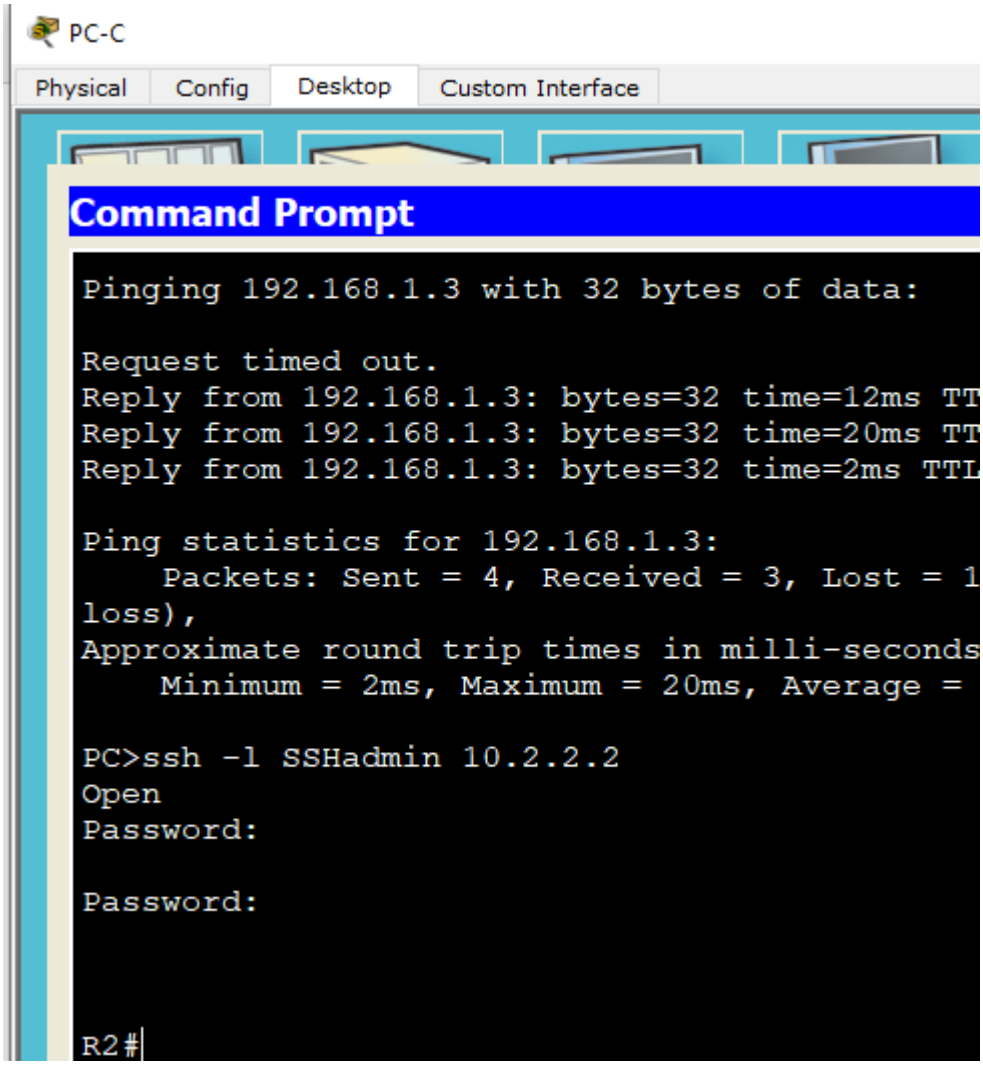
Step 1: From internal PC-C, ping the external PC-A server.



Step 2: From internal PC-C, SSH to the R2 S0/0/1 interface.

a. From the PC-C command prompt, SSH to R2 at 10.2.2.2.

ssh -l SSHadmin 10.2.2.2



b. While the SSH session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.

R3#show policy-map type inspect zone-pair sessions

R3

Physical | Config | CLI

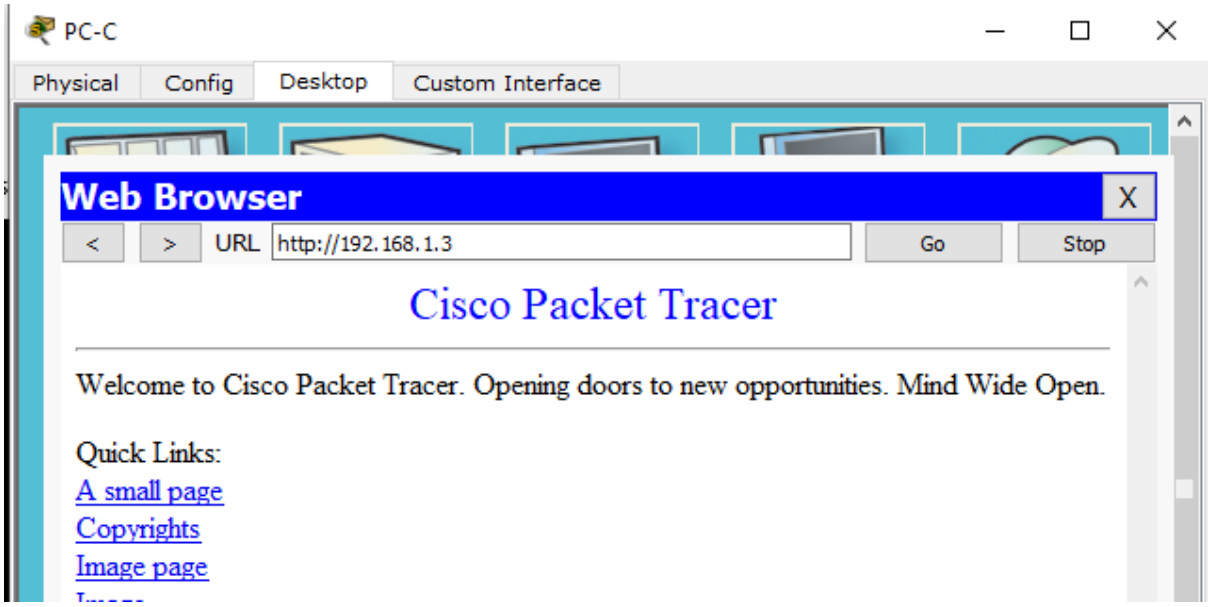
IOS Command Line Interface

```
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 1
        Established Sessions
         Session 199069584
(192.168.3.3:1027)=>(10.2.2.2:22) tcp SIS_OPEN/TCP_ESTAB
          Created 00:01:54, Last heard  00:00:42
          Bytes sent (initiator:responder) [1484:1540]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
R3#
```

Step 3: From PC-C, exit the SSH session on R2 and close the command prompt window.

```
PC>ssh -l SSHadmin 10.2.2.2
Open
Password:

Password:



R2#
R2#exit

[Connection to 10.2.2.2 closed by foreign host]
PC>
```

Step 4: From internal PC-C, open a web browser to the PC-A server web page.



While the HTTP session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.

R3#show policy-map type inspect zone-pair sessions

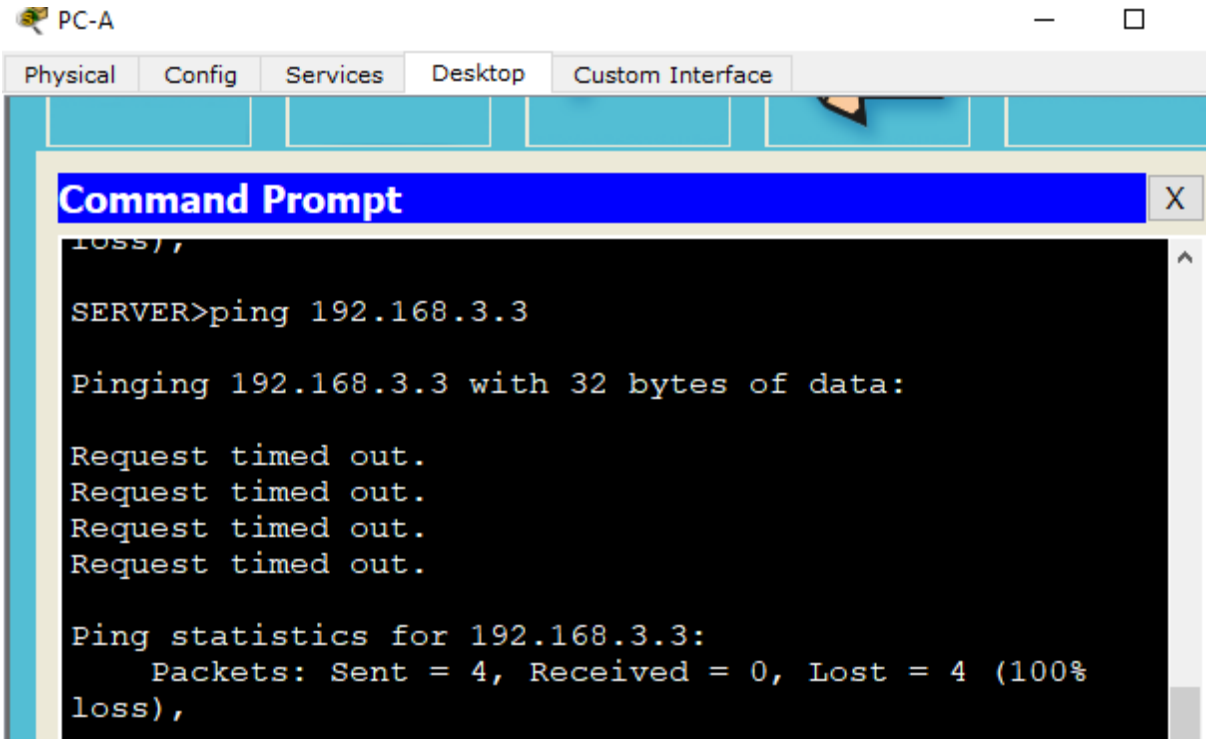
```
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 2
        Established Sessions
         Session 195321336
(192.168.3.3:1030)=>(192.168.1.3:80) tcp
SIS_OPEN/TCP_ESTAB
          Created 00:00:04, Last heard  00:00:04
          Bytes sent (initiator:responder) [284:575]
         Session 195318456
(192.168.3.3:1031)=>(192.168.1.3:80) tcp
SIS_OPEN/TCP_ESTAB
          Created 00:00:02, Last heard  00:00:02
```

Step 5: Close the browser on PC-C.

Part 7: Test Firewall Functionality from OUT-ZONE to IN-ZONE

Step 1: From the PC-A server command prompt, ping PC-C.

From the PC-A command prompt, ping PC-C at 192.168.3.3. The ping should fail.



Step 2: From R2, ping PC-C.

From R2, ping PC-C at 192.168.3.3. The ping should fail.

```
R2#ping 192.168.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is
2 seconds:
.....
Success rate is 0 percent (0/5)
```